

TECHNICAL MEMORANDUM

Date:	November 8, 2021
To:	Jessica Law and Erica Bishop – Sacramento Water Forum
From:	Matthew Weber and Chris Hammersmark – cbec eco engineering
Project:	17-1005 – LAR Current Condition DEM and 2D Model Development Project
Subject:	Salmonid Rearing Habitat Estimation

1 OBJECTIVE

To estimate the amount of rearing habitat for juvenile salmonids (steelhead and Chinook salmon) within the lower American River (LAR) across a wide range of flow conditions using the 2017 digital elevation model and HEC-RAS two-dimensional (2D) hydrodynamic model.

2 INTRODUCTION

A 2D HEC-RAS model developed for this Project by cbec (cbec 2019a and cbec 2019b) for the Sacramento Water Forum (Water Forum) and the Sacramento Area Flood Control Agency (SAFCA) was used to estimate the available salmonid habitat within LAR. The 2D model simulates steady flow conditions using 2017 topo-bathymetric LiDAR and sonar surveys to represent 2017 topography, bathymetry, and roughness conditions (Quantum Spatial 2018 and cbec 2018).

2.1 SALMONID LIFE STAGES

The two species of interest are the Central Valley Fall-run Chinook Salmon (*Oncorhynchus tshawytscha*), listed as a Species of Concern under the Endangered Species Act, and the Central Valley Steelhead (*O. mykiss*), which is a Federally Threatened Species. Both species spawn and rear in LAR with the periodicity of their lifestages shown in Figure 1. LAR fall-run Chinook salmon rear from early January to early June, with the late January through early April period typically having the most juveniles rearing in LAR. LAR steelhead juvenile rearing occurs year-round, since one life strategy of steelhead is to remain in the natal stream year-round and/or delay migration to the ocean for their adult life stage.

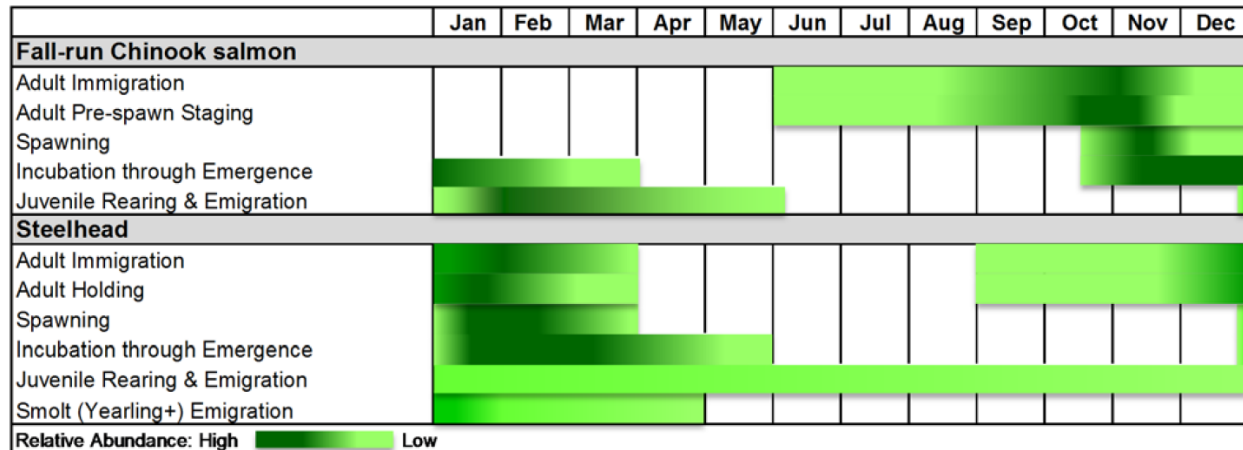


Figure 1. Lifestage periodicities for fall-run Chinook salmon and steelhead in the lower American River. Shading indicates intensity of the particular lifestage (copied directly from Bratovich et al., 2017).

3 METHODS

3.1 FLOWS MODELED

Steady inflow simulations were modeled in HEC-RAS 2D using existing hydraulic models for baseflow (i.e., 500 cfs) up to the peak design discharge for the LAR (i.e., 160000 cfs). The ecological flow model (cbec 2019a) was used for flows below 20,000 cfs and the flood flow model (cbec 2019b) was used for flows greater than 20,000 cfs. Flows modeled include: 500, 800, 1000, 1250, 1500, 1750, 2000, 2750, 3500, 4500, 5500, 6500, 7500, 8500, 10000, 15000, 30000, 45000, 70000, 115000, 160000 cfs. Since the ecologically relevant flows occur in the 500 to 20,000 cfs range, most of the simulations fall within that range.

3.2 HABITAT SUITABILITY INDICES

Salmonid juvenile rearing hydraulic HSI (shown in Figure 2) were taken from curves developed as part of the Voluntary Settlement Agreement process¹ and reflect juvenile salmonid observations from 11 northern California rivers. In addition to the hydraulic model outputs (i.e., depth and velocity), a cover map was developed for juvenile rearing habitat calculations. Table 1 shows the cover HSI values used for juvenile rearing habitat.

3.3 WEIGHTED USABLE AREA CALCULATION

Depth and velocity outputs from the two-dimensional model were used to calculate hydraulic habitat suitability index (hHSI) values for juvenile rearing and spawning habitat (see equation 1 below). For juvenile rearing, an additional step was undertaken to factor in cover habitat suitability (i.e., CHSI) to calculate a combined habitat suitability index (cHSI, see equation 2 below). Finally, the raster cell HSI values can be converted to a weighted usable area (WUA) to calculate a representative quantity of ideal

¹ The depth and velocity rearing suitability values used in this study were developed for the California State Water Resources Control Board Voluntary Settlement Agreement and were agreed upon by a committee of interested stakeholders as part of the Voluntary Settlement Agreement process during a meeting held November 9, 2017.

habitat by multiplying the HSI value within a cell by the cell area and summing all the resulting values (see equation 3 below). The equations are shown below,

$$hHSI_{cell} = \sqrt{VHSI \times DHSI} \quad (1)$$

$$cHSI_{cell} = \sqrt{VHSI \times DHSI} * CHSI \quad (2)$$

$$WUA = \sum(HSI_{cell} \times Area_{cell}) \quad (3)$$

where hHSI, VHSI, DHSI, CHSI, and CHSI are the hydraulic, velocity, depth, combined, and cover habitat suitability indices, respectively. Equation 1 represents the geometric mean approach for calculating the habitat suitability of the hydraulic outputs (i.e., depth and velocity). Equation 2 represents a hybrid approach where the geometric mean is used for the hydraulic variables and cover habitat suitability is treated as a factor (i.e., multiplicative approach). The VHSI and DHSI values used are shown in Figure 2 and the CHSI values are shown in Table 1.

Table 1. Juvenile salmonid cover habitat suitability indices (CHSI)

Type of Cover Habitat	Distance to Habitat	Cover HSI (CHSI)
2000 cfs channel	Inside area	0.2
Woody Vegetation	0 to 3 ft	1
Woody Vegetation	3 to 6 ft	0.85
Woody Vegetation	6 to 9 ft	0.35
Woody Vegetation	9 to 12 ft	0.25
Herbaceous Vegetation	inside area	0.5
Sparse shrub/sapling	inside area	0.5
Unvegetated floodplain	inside area	0.2

4 RESULTS

Salmonid rearing HSI and WUA results were calculated using the geometric mean equation for the hydraulic habitat suitability variables (i.e., hHSI, Equation 1) and then factoring in cover habitat suitability as a multiplicative factor (i.e., cHSI, Equation 2). Figure 3 shows that when analyzing only the hydraulic variables, there is a large amount of habitat at low flow, but after accounting for cover habitat suitability, those values drop significantly. This is typical, as baseflow channels typically contain large areas of shallow slow-moving water, but do not typically contain large areas with suitable cover.

Figure 4 shows bar graphs of the WUA results for hHSI by discharge broken down into five bins of habitat quality. The highest quality bin (blue, HSI 0.81 – 1.0) represents areas with the most ideal hydraulic conditions. Figure 5 shows these same bar graphs while factoring in cover habitat (cHSI). Once cover habitat is incorporated, the quality of the habitat is significantly decreased for the lower flows until the water begins interacting with the floodplain and bank vegetation.

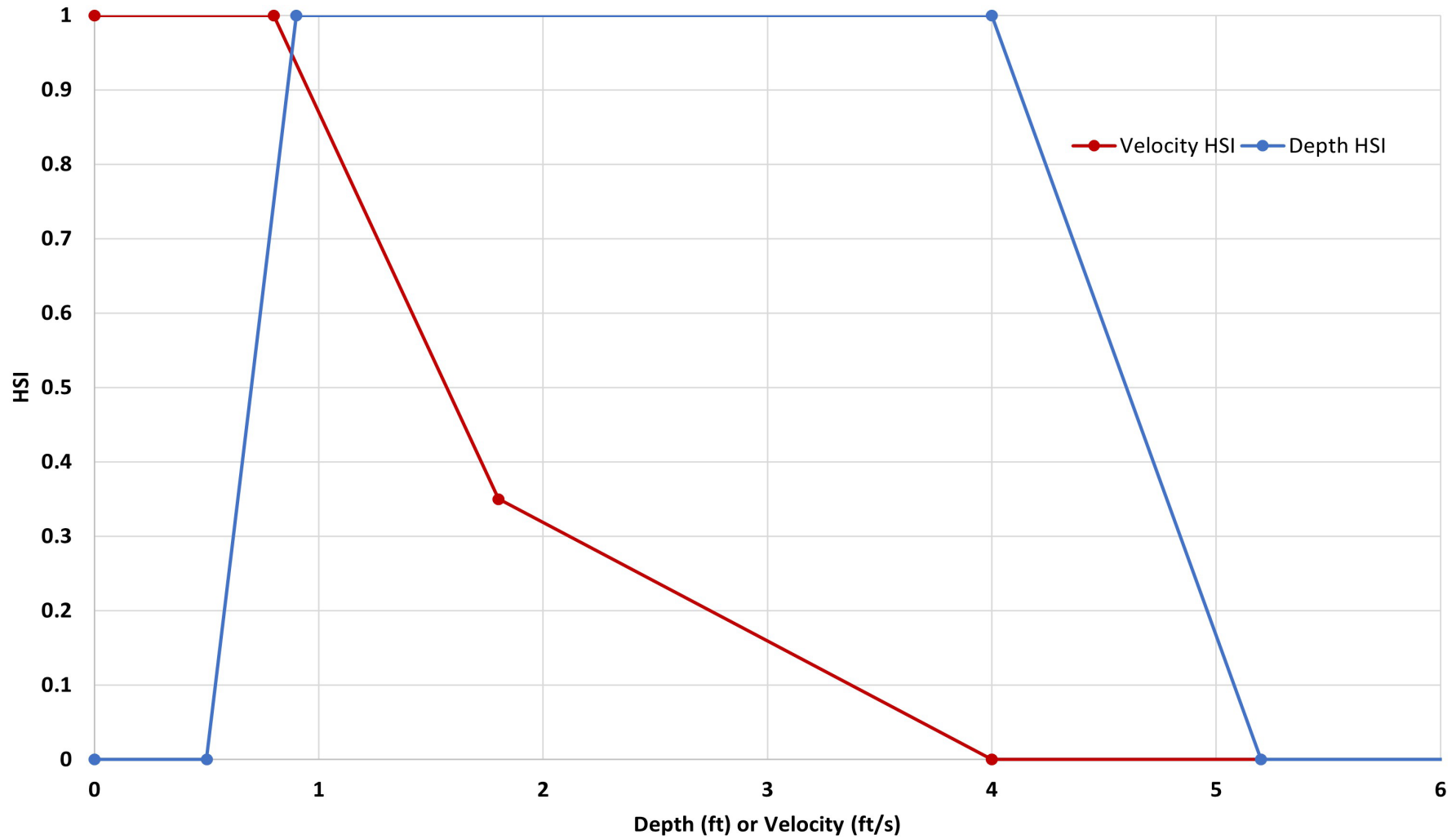
To show the spatial variability of habitat availability (as expressed as WUA) across the model domain, the longitudinal profile of the LAR was divided into 10 tiles that each span a range of 2 – 2.25 river miles from the confluence with the Sacramento River to Nimbus Dam. Longitudinal WUA profiles are shown in Figure 6 (hHSI, without cover) and Figure 7 (cHSI, with cover) for six flows from 500 cfs to 30,000 cfs. These results both show that WUA values peak towards the middle reaches of the river, but the peak in values is more pronounced in the results with cover (Figure 7). This is the Riverbend Reach, where the river is widest, multithreaded, and has more significant floodplain interaction with vegetation.

5 SUMMARY

This analysis provides the best-to-date analysis of rearing habitat availability within LAR with a comprehensive spatial area and range of flows analyzed. The results highlight the importance of frequently inundated riparian habitat for juvenile salmonids (especially outside of the Riverbend Reach). This rearing habitat information has already been used to help inform habitat restoration efforts for the Water Forum Rearing Sites Identification and Prioritization Project (cbec 2020), the CVPIA Science Integration Team's Coarse Resolution Model, the Sacramento County Regional Parks Natural Resources Management Plan, and the American River Common Feature Project.

6 REFERENCES

- Bratovich, P., Weaver, J., Addley, C., and Hammersmark, C. 2017. Lower American River Biological Rationale, Development and Performance of the Modified Flow Management Standard. Exhibit ARWA-702, November 29, 2017.
- cbec, inc., 2018. LAR Current Condition DEM and 2D Model Development Project – Current Conditions DEM Report. Prepared for The Sacramento Water Forum and The Sacramento Area Flood Control Agency.
- cbec, inc., 2019a. Lower American River Ecological Flow Hydrodynamic Modeling Report. Prepared for The Water Forum and the Sacramento Area Flood Control Agency.
- cbec, inc., 2019b. Lower American River Flood Flow Hydrodynamic Modeling Report. Prepared for The Water Forum and the Sacramento Area Flood Control Agency.
- cbec, inc., 2020. Lower American River Salmonid Rearing Habitat Enhancement Project Identification and Prioritization Report. Prepared for the Sacramento Water Forum and the U.S. Fish and Wildlife Service.
- Quantum Spatial, 2018. American River, California Topobathymetric LiDAR & Digital Imagery Technical Data Report. Prepared for cbec inc. eco engineering.



Notes:



LAR Current Condition DEM and 2D Model Development Project
Salmonid rearing DHSI and VHSI

Project No. 17-1005

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Figure 2

